Al-Balqa Applied University

Note: Write all the required equations (without simplification) to solve the following problems, then find the final answer.

V=2A

Prob. 1: *Consider the circuit shown in Fig. 1,*

1. The open circuit voltage between *a-b* terminals (V_o), when the load impedance is disconnected, is:

(8-j4)//(5+10j) $I_5 * 5\Omega = V_o = 23,4 - 29,2j$

2. The short circuit current (I_{sc}), when *a-b* terminals are shorted is:

$I_{sc} = \frac{V_2}{Z_2} \Rightarrow \frac{9,6 + 48j}{10j} = 4,8 - 9,6j$

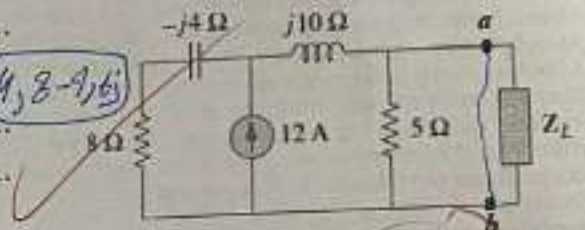


Fig.1

3. If the load impedance $Z_L = 8+j6\,\Omega$, The load current is:

$V_L = V_5 = 20 - 20j$

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Prob. 2: *Consider the circuit shown in Fig. 2 and using mesh analysis method,*

1. Write the two mesh equations, do not simplify:

$I_1 \Rightarrow -40\angle 30 + 10j I_1 - 20j(I_1 - I_2) = 0$

$I_2 \Rightarrow 40 I_2 + 50 - 20j(I_2 - I_1) = 0$

2. The current I_1 is: $-0,428 + 4,678j$

3. The current I_2 is: $0,725 + 0,607j$

4. The capacitor current, I_C, is: $21,42 + 24,26j$

2

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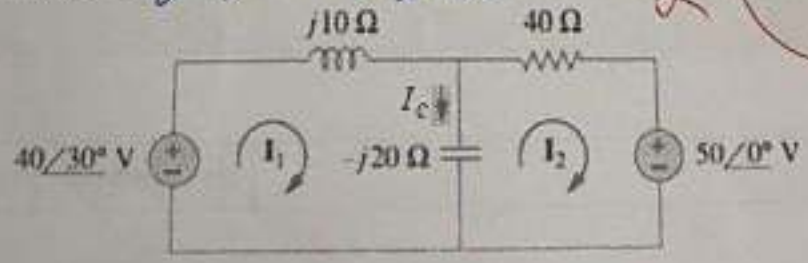


Fig 2

Consider the circuit shown in Fig. 3 .

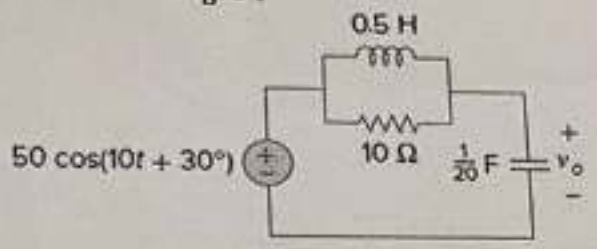


Figure 3

1. The equivalent admittance across the terminals of the source is:

$Y = \frac{1}{Z} \rightarrow 0,25 - 0,25j$

2.The voltage (V0) is:

$V_o = I \cdot Z_{eq} \rightarrow -9,15 - 34,15j$

3.The current through 10Ω resistor in phasor domain is:

$I_{10} = \frac{V_o}{Z} \rightarrow 5,24 + 5,91j$

4.The total current lags the voltage of the voltage source by :

45 degrees

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